MEDUSA FTP AUTHENTICATION TESTING

Lab Report — Kali Linux and Metasploitable2

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1. Objective

To learn basic Medusa usage for a controlled FTP authentication check, identify the target service with Nmap, connect using an FTP client, and inspect FTP packets in Wireshark.

2. Lab Environment

- Tester: Kali Linux, IP 192.168.10.132 (shown in the capture).
- Target: Metasploitable2, IP 192.168.10.130.
- Tools: Nmap, Medusa v2.3, FTP client, Wireshark.
- Scope: Intentionally vulnerable lab machine only.

3. Method and Commands

3.1 Nmap service identification

Command: nmap -sV -p 21 192.168.10.130

Result: TCP port 21 was open and identified as FTP running vsftpd 2.3.4.

Command: nmap -sC -sV -p 21 192.168.10.130

The script output reported anonymous FTP login allowed and stated that control and data connections were plaintext.

3.2 Medusa module inspection

Commands: medusa -d | medusa -M ftp -q

Medusa v2.3 listed modules including FTP, SSH, Telnet, MySQL, PostgreSQL, VNC, SMB, HTTP and others. The FTP module help described NORMAL, EXPLICIT and IMPLICIT FTP/TLS modes.

3.3 FTP authentication check

Command used: medusa -h 192.168.10.130 -u msfadmin -p 'msfadmin' -M ftp

Result: Medusa printed ACCOUNT FOUND for the FTP module and marked the supplied msfadmin credential as SUCCESS. This was a single-credential check, not a password-list or high-volume test.

3.4 Manual FTP login and directory browsing

Command: ftp 192.168.10.130

The FTP client returned “230 Login successful.” The ls command showed a vulnerable directory. Within it, folders included mysql-ssl, samba, tikiwiki and twiki20030201. The Samba-related directory contained 3.0.20, 3.0.6 and deps; the 3.0.20 directory contained debs, jduck-build-stuff.tar.gz and samba-3.0.20.tar.gz.

3.5 Wireshark inspection

Display filter: ip.addr == 192.168.10.130 && tcp.port == 21

The capture showed traffic between 192.168.10.132 and 192.168.10.130, including FTP USER/PASS commands and the server’s “230 Login successful” response. This matches the scan’s finding that FTP traffic was plaintext.

4. Results

Test	Observed result
Target/service	Port 21/tcp open; vsftpd 2.3.4 detected.
Anonymous FTP	Nmap script output reported anonymous login allowed.
Medusa	Supplied msfadmin credential accepted by FTP module.
Manual login	FTP client returned “230 Login successful.”
Directory listing	Directories and archive files were visible to the logged-in session.
Packet capture	Wireshark displayed FTP commands and responses in plaintext.

5. Security Observations

- FTP without TLS may expose usernames, passwords and commands to anyone able to capture the traffic.
- Weak or reused credentials can increase the risk of unauthorized access.
- Disable anonymous FTP unless it is explicitly needed and appropriately restricted.
- Prefer SFTP or properly configured FTPS over plaintext FTP.
- Use unique strong passwords, restrict service exposure, and monitor repeated authentication failures.

6. Conclusion

The lab demonstrated a basic Medusa FTP authentication check against Metasploitable2. Nmap identified vsftpd 2.3.4 on port 21, Medusa validated the supplied lab credential, manual FTP login succeeded, and Wireshark showed the FTP exchange. The results demonstrate the risks of plaintext FTP and weak or reused credentials. Testing should remain limited to systems for which explicit authorization has been granted.

7. Evidence Screenshots

Figure 1. Nmap service scan of the Metasploitable2 target.

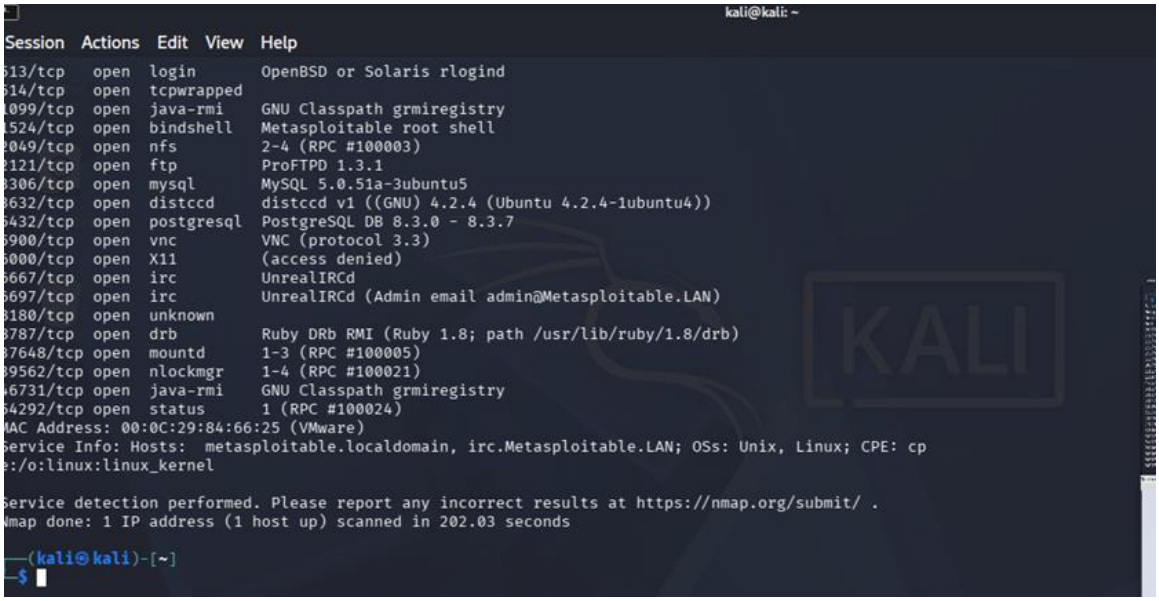


Figure 2. Nmap FTP version detection for port 21.

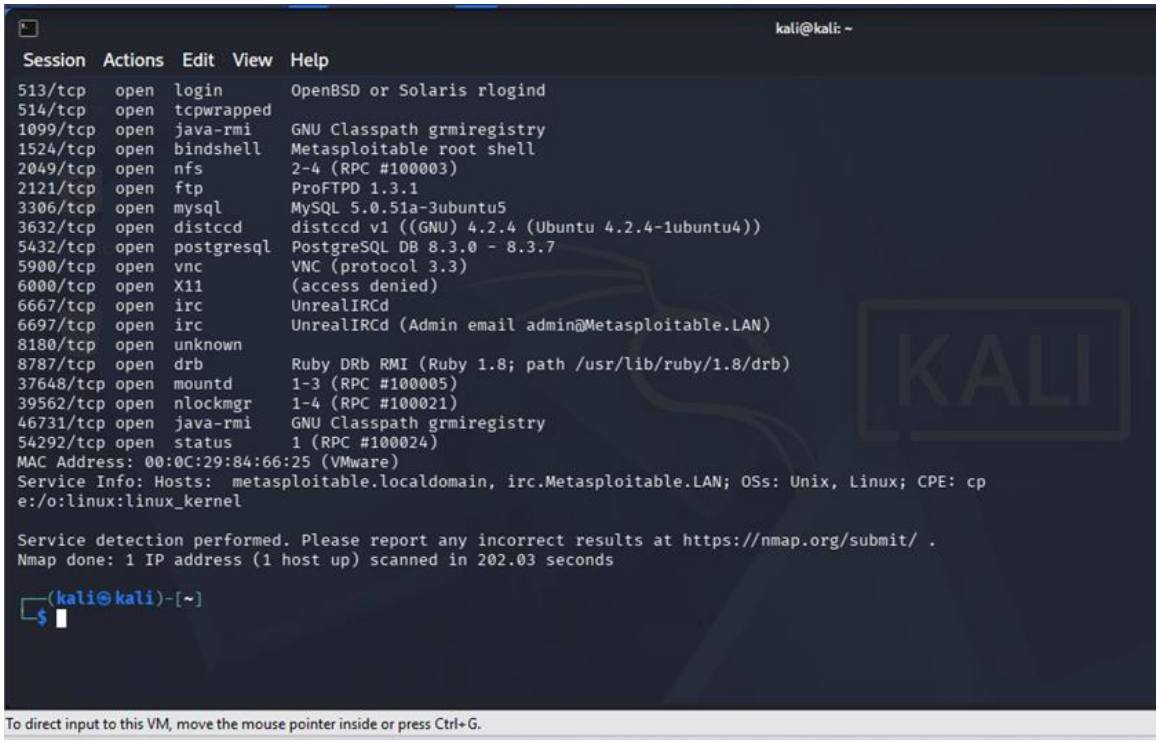


Figure 3. Nmap default-script scan and FTP configuration.

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV -p 21 192.168.10.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-31 01:36 -0400
Nmap scan report for 192.168.10.130
Host is up (0.0010s latency).

PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.3.4
MAC Address: 00:0C:29:84:66:25 (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.00 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 4. Medusa FTP module options.

```
Nmap done: 1 IP address (1 host up) scanned in 1.00 seconds

┌──(kali㉿kali)-[~]
└─$ nmap -sC -sV -p 21 192.168.10.130
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-31 01:37 -0400
Nmap scan report for 192.168.10.130
Host is up (0.00087s latency).

PORT   STATE SERVICE VERSION
21/tcp open  ftp     vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to 192.168.10.132
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      vsFTPd 2.3.4 - secure, fast, stable
|_End of status
MAC Address: 00:0C:29:84:66:25 (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.36 seconds

┌──(kali㉿kali)-[~]
└─$
```

Figure 5. Medusa reports a successful FTP credential check.

```
┌──(kali㉿kali)-[~]
└─$ medusa -M ftp -q
Medusa v2.3 [http://www.foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>

ftp.mod (2.1) pMonkey <pmonkey@foofus.net> :: Brute force module for FTP/FTPS sessions

Available module options:
MODE:? (NORMAL*, EXPLICIT, IMPLICIT)

  EXPLICIT: AUTH TLS Mode as defined in RFC 4217
     Explicit FTPS (FTP/SSL) connects to a FTP service in the clear. Prior to
     sending any credentials, however, an "AUTH TLS" command is issued and a
     SSL session is negotiated.

  IMPLICIT: FTP over SSL (990/tcp)
     Implicit FTPS requires a SSL handshake to be performed before any FTP
     commands are sent. This service typically resides on tcp/990. If the user
     specifies this option or uses the "-n" (SSL) option, the module will
     default to this mode and tcp/990.

  NORMAL
     The default behaviour if no MODE is specified. Authentication is attempted
     in the clear. If the server requests encryption for the given user,
     Explicit FTPS is utilized.


Example Usage:
    medusa -M ftp -h host -u username -p password
    medusa -M ftp -s -h host -u username -p password
    medusa -M ftp -m MODE:EXPLICIT -h host -u username -p password
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Figure 6. Medusa authentication test output.

```
2026-10-31 01:39:30 ACCOUNT FOUND: [ftp] Host: 192.168.10.130 User: msfadmin Password: msfadmin [SUCCESS]

┌──(kali㉿kali)-[~]
└─$ medusa -h 192.168.10.130 -u msfadmin -p msfadmin -M ftp
Medusa v2.3 [http://www.foofus.net] (C) JoMo-Kun / Foofus Networks <jmk@foofus.net>

2026-10-31 01:41:31 ACCOUNT CHECK: [ftp] Host: 192.168.10.130 (1 of 1, 0 complete) User: msfadmin (1 of 1, 0 complete) Password: msfadmin (1 of 1 complete)
2026-10-31 01:41:31 ACCOUNT FOUND: [ftp] Host: 192.168.10.130 User: msfadmin Password: msfadmin [SUCCESS]
```

Figure 7. Wireshark capture filtered to FTP control traffic.

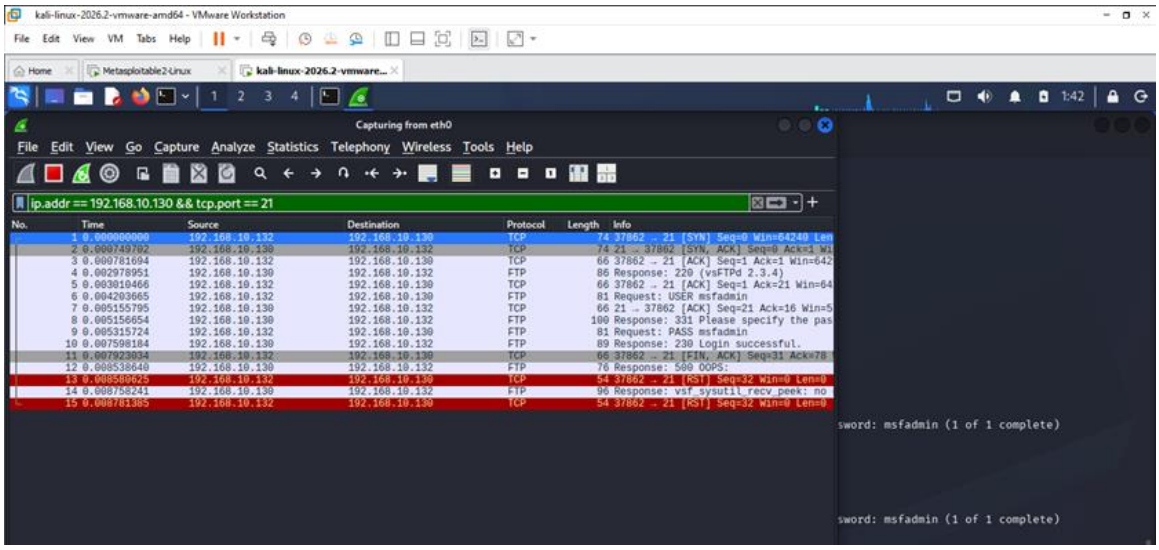
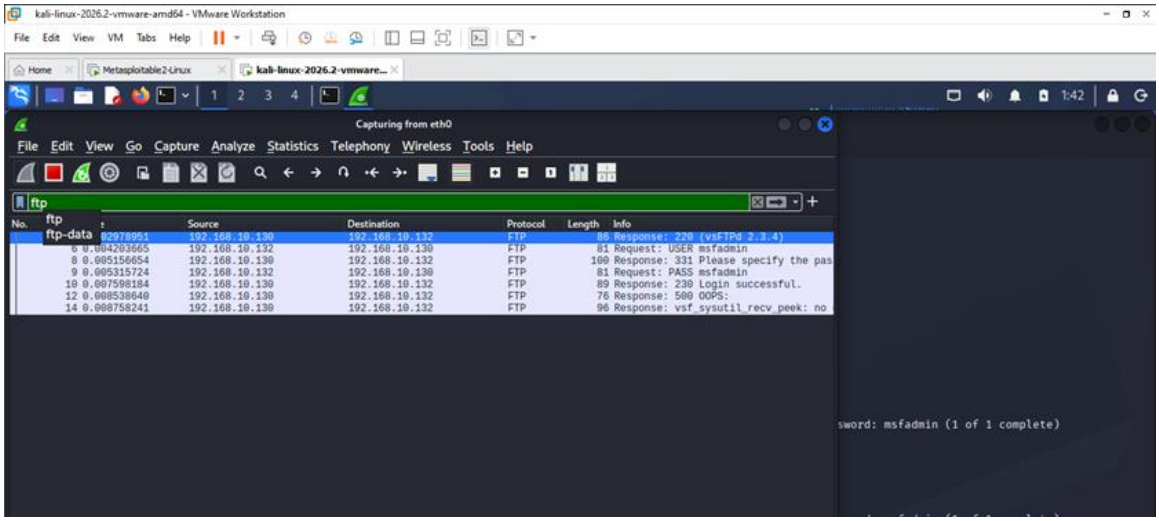


Figure 8. Wireshark FTP commands and server responses.



Appendix: Key Commands

nmap -sV -p 21 192.168.10.130

nmap -sC -sV -p 21 192.168.10.130

medusa -d

medusa -M ftp -q

medusa -h 192.168.10.130 -u msfadmin -p 'msfadmin' -M ftp

ftp 192.168.10.130

Wireshark filter: ip.addr == 192.168.10.130 && tcp.port == 21